

Stony Brook University  
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## ASSIGNMENT 2: Budget Sets, Preferences & Utility

**Due: February 18, in class.**

INSTRUCTIONS: Show all your work or points will be deducted.

1. Violet is preparing for exams in economics and sociology. She has time to study 40 pages of economics and 30 pages of sociology. In the same amount of time she could also read 30 pages of economics and 60 pages of sociology.
  - (a) Assuming that the number of pages per hour that she can read of either subject does not depend on how she allocates her time, how many pages of sociology could she read if she decided to spend all of her time on sociology and none on economics? *Hint: you have two points on her budget line, so you should be able to determine the entire line.*
  - (b) How many pages of economics could she read if she decided to spend all of her time reading economics?
  - (c) Draw Violet's budget line. Be sure to label all relevant features (intercepts and slope).
  - (d) Suppose now that Violet has a friend, Brenda, who took in the previous term the same Economics class that Violet is taking. However, from the topics that Violet has to study, Brenda only remembers the first few chapters. Thus, Brenda can only help with the first 25 pages that Violet has to study. For these first 25 pages, with the help of Brenda, Violet has time to study 10 pages of economics and 140 pages of sociology, whereas in the same amount of time she could also read 20 pages of economics and 130 pages of sociology. How many pages of economics can Violet now read if she decided to spend all of her time reading economics? How many of Sociology?
  - (e) Draw Violet's new budget line after she receives help from Brenda. Be sure to label all relevant features, like intercepts slopes or *kinks*.
2. On the planet of Mongo, emperor *Ming the Merciless* forces his subjects to go through a rather contrived market exchange procedure: Mongo has two kinds of currencies, **blue money** and **red money**. Every commodity has two prices—a **red-money** price and a **blue-money** price. Every Mongoan has two incomes—a **red** income and a **blue** income.  
 In order to buy an object, a Mongoan has to pay that object's **red-money** price in **red** money and its **blue-money** price in **blue** money. It is forbidden to trade one kind of money for the other, and this prohibition is strictly enforced by *Ming's* ruthless enforcers.
  - There are just two consumer goods on Mongo, ambrosia and fuel. All Mongoans prefer more of each good to less.
  - The **blue** prices are **1 bcu** (bcu stands for blue currency unit.) per unit of ambrosia and **1 bcu** per unit of fuel.
  - The **red** prices are **2 rcus** (red currency units) per unit of ambrosia and **6 rcus** per unit of fuel.

- (a) Draw the **red budget** (with red ink) and the **blue budget** (with blue ink) for a Mongoan named Barin whose **blue income is 10** and his **red income is 30**. Shade in the “budget set” containing all of the commodity bundles that Barin can afford, given his two budget constraints. Remember, Barin has to have enough blue money and enough red money to pay both the blue-money cost and the red-money cost of a bundle of goods.
- (b) Another Mongoan, Aura, faces the same prices that Barin faces and has the **same red income** as Barin, but Aura has a **blue income of 20**. Explain how it is that Aura will not spend its entire blue income no matter what her tastes may be. *Hint: Draw Aura’s budget lines.*
- (c) A group of radical economic reformers<sup>1</sup> arrive to Mongo and believe that the currency rules are unfair. “Why should everyone have to pay two prices for everything?” they ask. They propose the following scheme. Mongo will continue to have two currencies, every good will have a blue price and a red price, and every Mongoan will have a blue income and a red income. But nobody has to pay two prices. Instead, everyone on Mongo must declare itself to be either a **Blue-Money Purchaser** (a “Blue”) or a **Red-Money Purchaser** (a “Red”) before he or she buys anything at all. **Blues** must make all their purchases in blue money at the blue prices, spending only their blue incomes. **Reds** must make all of their purchases in red money, spending only their red incomes.
- Suppose that Barin has the same income after this reform, and that prices do not change. Before declaring which kind of purchaser he will be, Barin contemplates the set of commodity bundles that he could afford by making one declaration or the other. Let us call a commodity bundle “attainable” if Barin can afford it by declaring himself to be a **Blue** and buying the bundle with blue money *or* if Barin can afford the bundle by declaring itself to be a **Red** and buying it with red money. Draw correctly labeled diagram in which you shade in all of the attainable bundles.
- (d) Are Mongoan budgets really so fanciful? Can you think of situations on Earth where people must simultaneously satisfy more than one budget constraint? Is money the only scarce resource that people use up?
3. Coach Steroid likes his players to be big, fast and obedient. If player *A* is better than player *B* in two of these three characteristics, then Coach Steroid prefers *A* to *B*, but if *B* is better than *A* in two of these three characteristics, then Steroid prefers *B* to *A*. Otherwise, Steroid is indifferent between them. Wilbur weighs 340 pounds, runs very slowly, and is fairly obedient. Harold weighs 240 pounds, runs very fast, and is very disobedient. Jerry weighs 150 pounds, runs at average speed, and is extremely obedient.
- (a) Does Steroid prefer Wilbur to Harold or vice versa?
- (b) Does Steroid prefer Harold to Jerry or vice versa?
- (c) Does Steroid prefer Wilbur to Jerry or vice versa?
- (d) Does Coach Steroid have transitive preferences?
- (e) After several losing seasons, Coach Steroid decides to change his way of judging players. According to his new preferences, Steroid prefers player *A* to player *B* if player *A* is better in all three of the characteristics that Steroid values, and he prefers *B* to *A* if player *B* is better at all three things. He is indifferent between *A* and *B* if they weigh the same, are equally fast, and are equally obedient. In all other cases, Coach Steroid simply says “*A* and *B* are not comparable.”

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<sup>1</sup>Led by Flash Gordon, Dale Arden and Dr. Hans Zarkov.

- (f) Are Coach Steroid's new preferences complete?
- (g) Are Coach Steroid's new preferences transitive?
4. Dale consumes only nuts and berries. Fortunately, he likes both goods. The consumption bundle where Dale consumes  $x_1$  units of nuts per week and  $x_2$  units of berries per week is written as  $(x_1, x_2)$ . The set of consumption bundles  $(x_1, x_2)$  such that Dale is indifferent between  $(x_1, x_2)$  and  $(1, 16)$  is the set of bundles such that  $x_1 \geq 0, x_2 \geq 0$ , and  $x_2 = 20 - 4\sqrt{x_1}$ . The set of bundles  $(x_1, x_2)$  such that  $(x_1, x_2) \sim (36, 0)$  is the set of bundles such that  $x_1 \geq 0, x_2 \geq 0$  and  $x_2 = 24 - 4\sqrt{x_1}$ .
- (a) Sketch the indifference curve that passes through the point  $(1, 16)$ . Use blue ink for this. Similarly, using red ink, draw the indifference curve that passes through the point  $(36, 0)$ .
- (b) Shade the set of commodity bundles that Dale weakly prefers to the bundle  $(1, 16)$  with blue. Similarly, shade with red the set of commodity bundles  $(x_1, x_2)$  such that Dale weakly prefers  $(36, 0)$  to these bundles. Is the set of bundles that Dale prefers to  $(1, 16)$  a convex set?
- (c) What is the slope of Dale's indifference curve at the points  $(9, 8)$ ,  $(4, 12)$ ,  $(9, 12)$  and  $(4, 16)$ ?
- (d) Do the indifference curves you have drawn for Dale exhibit diminishing marginal rate of substitution?
- (e) Does Dale have convex preferences?
5. In this exercise, you will explore a preference relation and the utility that represents it that incorporates a *satiation point*. A satiation point is a bundle that provides complete "bliss" for the consumer. Suppose the bliss point is  $(b_1^*, b_2^*)$ . The ranking of all other bundles is determined by how close they are to  $(x^*, y^*)$ . Thus we have

$$(x_1, x_2) \succeq (y_1, y_2) \Leftrightarrow \sqrt{(x_1 - b_1^*)^2 + (x_2 - b_2^*)^2} \leq \sqrt{(y_1 - b_1^*)^2 + (y_2 - b_2^*)^2}$$

- (a) Strawberry Shortcake likes to spend some time studying and some time dating. Her preferences can be represented by the aforementioned utility function, whose indifference curves between hours per week spent studying and hours per week spent dating are concentric circles around her favorite combination, which is 20 hours of studying and 15 hours of dating per week. Surely, the closer she is to her favorite combination, the happier she is.
- (b) Graph some indifference curves for this preference relation. (Hint: A circle with center  $(b_1^*, b_2^*)$  and radius  $k$  is defined by the equation  $(x_1 - b_1^*)^2 + (x_2 - b_2^*)^2 = k^2$ . You may also want to use the monotonic transformation  $f(u) = u^2$  defined for any positive real number  $u$ .)
- (c) Suppose that Strawberry is currently studying 25 hours a week and dating 3 hours a week. Would she prefer to be studying 30 hours a week and dating 8 hours a week? Justify your answer.
- (d) Compute the slope of the indifference curves at the combinations of studying 25 hours a week/dating 3 hours a week and 30 hours a week/dating 8 hours a week.
6. Bert has preferences represented by the utility function  $u(a, b) = ab/100$ , where  $a$  is the number of ounces of animal crackers that he consumes and  $b$  is the number of ounces of broccoli that he consumes.
- (a) Sketch the set of points that Bert finds indifferent to having 8 ounces of animal crackers and 2 ounces of broccoli. Also sketch the locus of points that Bert finds indifferent to having 6 ounces of animal crackers and 4 ounces of broccoli.

- (b) Ernie has preferences represented by the utility function  $v(a, b) = 1000a^2b^2$ , where  $a$  is the number of ounces of animal crackers that he consumes and  $b$  is the number of ounces of broccoli that he consumes. Sketch the set of points that Ernie finds indifferent to having 8 ounces of animal crackers and 2 ounces of broccoli. Also sketch the locus of points that Ernie finds indifferent to having 6 ounces of animal crackers and 4 ounces of broccoli.
- (c) Are Bert's preferences convex? Are Ernie's?
- (d) What can you say about the difference between the indifference curves you drew for Bert and those you drew for Ernie?
- (e) How could you tell this was going to happen without having to draw the curves? Justify your answer.